

CaCo: Attributed Network Anomaly Detection via Canonical Correlation Analysis

Ruidong Wang , Fengbin Zhang , Xunhua Huang , Chongrui Tian, Liang Xi ,
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Abstract—Capturing the complex interaction between the node attribute and the network structure is important for attributed network embedding and anomaly detection. However, there are few methods to explicitly model the correlation between these two views of the node attribute and the network structure. In this article, we propose an attributed network anomaly detection (CaCo) method based on the canonical correlation analysis, which assumes that there should be a strong correlation between the attribute and structure features of normal nodes, and a weak correlation one between those abnormal nodes, in the attributed networks. Consequently, a joint learning mechanism is designed in CaCo to explicitly measure the correlation between two views in the latent space. Specifically, the backbone of a weight-sharing graph convolutional network is employed to encode the node feature from two views of attribute and structure in the latent space, respectively. Then, a Kullback–Leibler divergence regularization is used to align the distributions of the two views. Finally, the parameters of CaCo are optimized by maximizing the correlation between attribute and structure features of normal nodes in the training phase, and anomalies can be detected by measuring the correlation between two views in the testing phase. Extensive experiments on six real-world datasets demonstrate the effectiveness of the proposed method compared to the state-of-the-art techniques.

Index Terms—Anomaly detection, attributed network embedding, canonical correlation analysis (CCA), graph neural networks (GNNs).

I. INTRODUCTION

ATTRIBUTED networks are typical graph data whose nodes represent entities and links indicate the relationship between entities [1], [2]. In recent years, a lot of researchers

Manuscript received 31 January 2023; accepted 28 March 2023. Date of publication 11 April 2023; date of current version 11 December 2023. This work was supported in part by the Key Project of Colleges and Universities of Henan Province under Grant 23A52002 and in part by the Science and Technology Innovation 2030–“New Generation of Artificial Intelligence” Major Project under Grant 2021ZD0111000. Paper no. TII-23-0324. (Corresponding authors: Fengbin Zhang; Haoyi Fan.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TII.2023.3266406>.

Digital Object Identifier 10.1109/TII.2023.3266406

have focused on graph analysis, such as node classification, link prediction, and anomaly detection [1], [3], [4]. Among them, attributed network anomaly detection is an important problem worth studying, which aims to detect nodes whose behavior is different from the majority of nodes. However, in attributed networks, anomalies are diverse owing to the complex interaction of network structure and node attributes. Due to the impact of network structure and node attributes of graph learning, traditional anomaly detection methods [5], [6] are unable to capture the complete information of the graph for attributed network anomaly detection.

Recently, various methods of attributed network anomaly detection have been proposed. Most of them focus on learning the node representation in the latent space, and detecting the anomalous nodes by the reconstruction error or traditional anomaly detection methods [7], [8]. Shallow methods [9], [10] utilize shallow mechanisms to capture the interactions between network structure and node attributes, which cannot fully address the challenges of high-dimensional feature computation. Graph neural networks (GNNs)-based methods [11], [12], [13], [14] leverage the advantages of the graph convolutional neural network on graph data to extract the low-dimensional node embedding and detect the anomalous nodes by one-class methods or reconstruction errors. Besides, some methods based on self-supervised learning (SSL) [15], [16], [17] learn data representations by maximizing the information from the two data augmentations of the inputs and using the data representations for downstream tasks.

Although existing methods have achieved considerable performance in attributed network anomaly detection, they still have some limitations. Some methods such as two-step-based methods [18] and graph autoencoder-based methods [11], [12] are designed to learn the graph embedding rather than detecting anomalous nodes, which is not the target of the anomaly detection task. Some methods focus on capturing the interactions between network structures and node attributes in two different latent spaces, rather than mining abnormal patterns in the attributed networks [10], [13]. Moreover, some SSL-based methods capture local and global relationships through data augmentation, which ignores the correlations between network structures and node attributes in the latent space [15], [17].

To overcome the abovementioned limitations, in this article, we propose a canonical correlation analysis (CCA)-based method for attributed network anomaly detection (CaCo). The motivation of CaCo is shown in Fig. 1. We assume that there

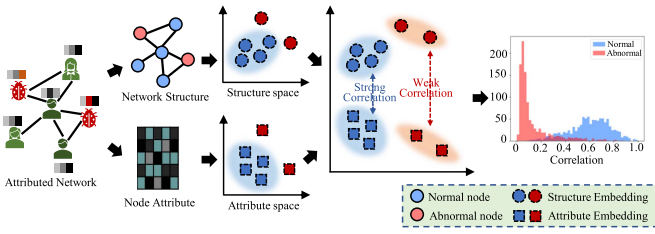


Fig. 1. Motivation of the proposed method. There should be a strong correlation between the attribute and structure feature for normal nodes, and weak correlation ones for abnormal nodes, in the attributed network.

should be a strong correlation between the attribute and structure features of normal nodes, and a weak correlation between those of abnormal nodes, in the attributed network. Therefore, CaCo is designed to learn a correlation measurement function between network structure distribution and node attribute distribution, and the abnormal nodes are detected based on the measured correlation scores. Specifically, in CaCo, the backbone of a weight-sharing graph convolutional network is employed to encode the node feature from two views of attribute and structure in the latent space, respectively. Then, a Kullback–Leibler (KL) divergence regularization is used to align the distributions of the two views. Finally, the parameters of CaCo are optimized by maximizing the correlation between attribute and structure features of normal nodes in the training phase, and anomalies can be detected by measuring the correlation between two views in the testing phase. Experimental results on six real-world datasets show that CaCo outperforms state-of-the-art techniques, which demonstrates the effectiveness of the proposed method.

The main contributions of this article are outlined as follows.

- 1) We redefine anomaly detection on attributed networks as a correlation metric problem between the attribute and structure of nodes and propose a framework of CCA-based attributed network anomaly detection, which explicitly models the correlation between these two views of node attribute and network structure.
- 2) We propose a correlation measurement method of the attribute and structure on attributed networks for anomaly detection, which aligns the distribution of attribute features and structure features of normal nodes in the latent space.
- 3) We conduct extensive experiments on multiple real-world datasets, and the results show that the proposed method achieves state of the art in the anomaly detection task of attributed networks.

II. RELATED WORKS

A. Traditional Anomaly Detection

Traditional anomaly detection is the task of detecting outliers for Euclidean structural data, such as tabular data or images. The existing methods can be divided into two main categories. First, one-class-classification-based methods [5], [6] attempt to learn a compact hyperplane or hypersphere to wrap the normal data, and data located outside the hyperplane or the hypersphere

are defined as outliers. Typical methods such as OC-SVM and SVDD [6] try to fit a hyperplane or hypersphere to separate the normal and abnormal data points. Second, the reconstruction-based approach assumes that anomaly data cannot be effectively reconstructed from a compressed low-dimensional vector space. For example, autoencoder-based methods [4], [7], [19] utilize the reconstruction error to detect the outliers. Although these methods have been successfully applied in anomaly detection, they do not generalize well to non-Euclidean graph data, which leaves the problem of anomaly detection on attributed networks still an open problem.

B. Attributed Network Anomaly Detection

Attributed network anomaly detection aims to find nodes whose patterns are significantly different from others. Existing graph anomaly detection methods can be divided into three categories [13]: 1) only structure- or attribute-based methods, 2) residual-analysis-based methods, and 3) GNN-based methods. The only structure- or attribute-based methods [18], [20] detect anomaly nodes by considering the structure information or node attribute information. These methods ignore the information of graphs, which makes them unable to achieve excellent performance. The residual-analysis-based methods [9], [10] utilize shallow mechanisms to capture the interaction between the network structure and node attributes to the local community. That make cannot capture the complete information of attributed networks. The GNN-based methods [13], [21], [22], [23] construct a graph autoencoder that uses both structure information and node attribute information by simultaneously reconstructing the attribute and structure and detects anomalies by one-class methods [6] or reconstruction errors. Besides, graph SSL-based methods [15], [24] transform graph data such as randomly masking certain information or randomly sampling subgraphs to generate input data and utilize GNNs to learn the graph embedding for anomaly detection. Although these methods have achieved significant applications in attributed network anomaly detection, they cannot achieve better performance by ignoring the correlation between network structure and node attributes or ignoring the effect of noise in the graph data.

C. Canonical Correlation Analysis

CCA [25], [26] is a method that aims to find the linear transformation for measuring the relationship between two vectors. Give two vectors $\mathbf{X}_1$ and $\mathbf{X}_2$, the correlation $\rho = \frac{\mathbf{a}^T \Sigma_{\mathbf{X}_1 \mathbf{X}_2} \mathbf{b}}{\sqrt{\mathbf{a}^T \Sigma_{\mathbf{X}_1 \mathbf{X}_1} \mathbf{a}} \sqrt{\mathbf{b}^T \Sigma_{\mathbf{X}_2 \mathbf{X}_2} \mathbf{b}}}$ is maximized by optimizing the objective

$$\max_{\mathbf{a}, \mathbf{b}} \mathbf{a}^T \Sigma_{\mathbf{X}_1 \mathbf{X}_2} \mathbf{b}, \text{ s.t. } \mathbf{a}^T \Sigma_{\mathbf{X}_1 \mathbf{X}_1} \mathbf{a} = \mathbf{b}^T \Sigma_{\mathbf{X}_2 \mathbf{X}_2} \mathbf{b} = \mathbf{I}. \quad (1)$$

The standard CCA and its variants are widely used for multiview learning [26], which uses linear projections to learn the feature embedding and utilizes the CCA objective to learn the correlation between different views. However, these models are linear in the sense. For learning the nonlinear embedding, Deep-CCA [25], [27] presents the use of deep neural networks to learn

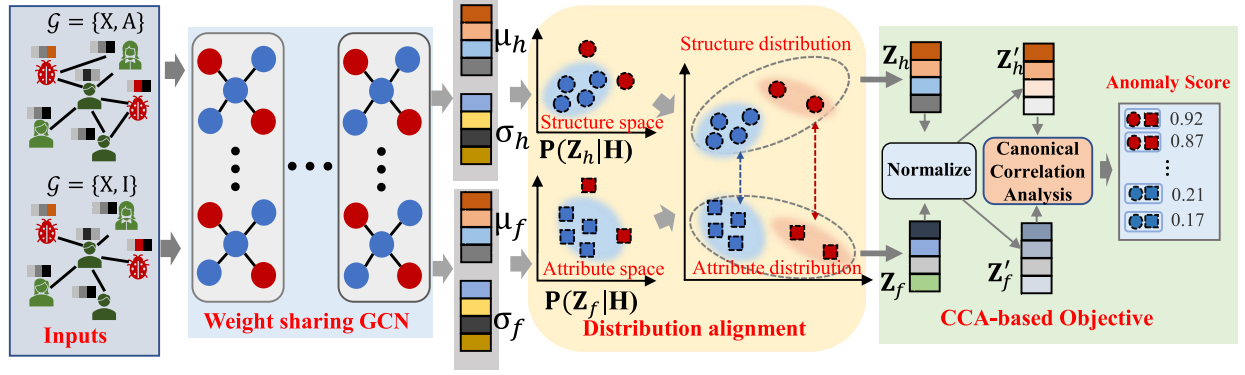


Fig. 2. Illustration of the proposed framework CaCo.

the nonlinear projections of different views. The use of exact decorrelation operations in those methods makes them achieve suboptimal optimization. Unlike these methods, Soft-CCA [28] considers the decorrelation constraint as a term of loss and optimizes it jointly with other terms, and the objective of Soft CCA is

$$\begin{aligned} \max_{\theta_1, \theta_2} & Tr(P_{\theta_1}^T(\mathbf{X}_1) P_{\theta_2}(\mathbf{X}_2)) \\ \text{s.t. } & P_{\theta_1}^T(\mathbf{X}_1) P_{\theta_1}(\mathbf{X}_1) = P_{\theta_2}^T(\mathbf{X}_2) P_{\theta_2}(\mathbf{X}_2) = \mathbf{I} \end{aligned} \quad (2)$$

where $\mathbf{I}$ is the identity matrix, and (2) can be rewritten as

$$\begin{aligned} \min_{\theta_1, \theta_2} & ||P_{\theta_1}(\mathbf{X}_1) - P_{\theta_2}(\mathbf{X}_2)||_F^2 \\ & + \lambda (\mathcal{L}_{\text{SDL}}(P_{\theta_1}(\mathbf{X}_1)) + \mathcal{L}_{\text{SDL}}(P_{\theta_2}(\mathbf{X}_2))) \end{aligned} \quad (3)$$

where P_{θ_1} and P_{θ_2} are the neural networks used to learn the representations of the two views. $||P_{\theta_1}(\mathbf{X}_1) - P_{\theta_2}(\mathbf{X}_2)||_F^2$ is used to maximize the correlation between the two views, and $\mathcal{L}_{\text{SDL}}$ is used to minimize the distance between $P_{\theta_i}(\mathbf{X}_i)$ and the identity matrix.

III. METHOD

In this section, we describe the proposed CaCo method, as shown in Fig. 2. CaCo consists of three parts: 1) a weight-sharing GCN, 2) a distribution alignment objective, and 3) a CCA-based objective. At first, in order to learn the distribution of the network structure and node attributes of attributed networks, we input the original graph $\mathcal{G} = \{\mathbf{A}, \mathbf{X}\}$ and the identity graph $\mathcal{G}' = \{\mathbf{A}, \mathbf{I}\}$ into the weight-sharing GCN, and output the mean and variance vectors of the network structure and node attributes. After that, to align the distributions of the network structure and node attributes, we use the KL divergence to pull the distributions of the two views to the same prior distribution. Finally, we sample the network structure embedding and node embedding from the learned distribution and maximize the correlation of normal nodes on the network structure distribution and node attribute distribution by using the CCA-based objective. Specifically, the anomaly scores of each node are defined as the correlations between network structure and node attributes, and the node

with a strong correlation between two views will be defined as a normal node.

A. Weight-Sharing Encoder

The weight-sharing encoder is designed to learn the correlation distribution of the network structure and node attributes. Specifically, CaCo has two different inputs: 1) the original graph $\mathcal{G} = \{\mathbf{A}, \mathbf{X}\}$ and 2) the identity graph $\mathcal{G}' = \{\mathbf{A}, \mathbf{I}\}$, which are utilized for learning the distribution of the network structure and node attributes.

To learn the structure information, the original graph $\mathcal{G} = \{\mathbf{A}, \mathbf{X}\}$ is input to the weight-sharing encoder to learn the network structure information of the attributed network

$$\text{GCN}(\mathbf{H}, \mathbf{A}|\mathbf{W}) = \varphi\left((\mathbf{D})^{-\frac{1}{2}}\mathbf{A}(\mathbf{D})^{-\frac{1}{2}}\mathbf{H}\mathbf{W}\right) \quad (4)$$

where $\mathbf{D}$ is the diagonal degree matrix of the network $\mathcal{G}$, $\mathbf{H} = \mathbf{X}$ is the node attributes, and $\mathbf{A}$ is the adjacency matrix of the attributed network.

To extract node semantic information, identity aggregation is designed to learn node attribute information. We utilize the identity graph $\mathcal{G}' = \{\mathbf{A}, \mathbf{I}\}$ input to the weight-sharing encoder

$$\text{GCN}(\mathbf{X}, \mathbf{I}|\mathbf{W}) = \varphi(\mathbf{I}\mathbf{X}\mathbf{W}) \quad (5)$$

where $\mathbf{I}$ is the identity graph, which makes the semantic information and network structure information of each node more similar by transferring the learned semantic information to all node features. $\mathbf{W}$ in (4) and (5) is the trainable shared weight. φ is the activation function, such as $\text{RELU}(\bullet) = \max(0, \bullet)$.

B. Distribution Alignment

After the weight-sharing GCN, the network structure distribution $q(\mathbf{Z}_h|\mathbf{H}, \mathbf{A})$ and node attribute distribution $q(\mathbf{Z}_f|\mathbf{X}, \mathbf{I})$ are learned by

$$q(\mathbf{Z}|\mathbf{X}, \mathbf{A}) = \prod_{i=0}^N q(\mathbf{z}_i|\mathbf{X}, \mathbf{A}) \quad (6)$$

$$q(\mathbf{z}_i|\mathbf{X}, \mathbf{A}) = \mathcal{N}(\mathbf{z}_i|\boldsymbol{\mu}_i, \text{diag}(\boldsymbol{\sigma}^2)) \quad (7)$$

where μ is the mean vectors and σ is the variance vectors. $\mathbf{Z}$ is the embedding sampled from the distributions of the two views. For the network structure distribution, μ_h and σ_h are the mean and variance vectors of the structure, respectively, and $\mathbf{Z}_h$ is the network embedding

$$\mu = \text{GCN}_{\mu}(\mathbf{H}, \mathbf{A}|\mathbf{W}) \quad (8)$$

$$\sigma = \text{GCN}_{\sigma}(\mathbf{H}, \mathbf{A}|\mathbf{W}) \quad (9)$$

μ_h and σ_h are the mean and variance vectors of the network structure distribution, respectively. Similarly, μ_f and σ_f are the mean and variance vectors of the node attributes, respectively.

Then, we align the learned distributions of the two views on attributed networks. Since it is difficult to align the two distributions directly, in this section, we utilize the KL divergence to pull the distribution $q(\mathbf{Z}_h|\mathbf{X}, \mathbf{A})$ and $q(\mathbf{Z}_f|\mathbf{X}, \mathbf{I})$ to the prior distribution p

$$\mathcal{L}_{kl} = -\text{KL}[q(\mathbf{Z}_h|\mathbf{X}, \mathbf{A})||p(\mathbf{Z}_h)] - \text{KL}[q(\mathbf{Z}_f|\mathbf{X}, \mathbf{I})||p(\mathbf{Z}_f)]. \quad (10)$$

In this article, we use the Gaussian distribution as the prior distribution p [22], [23].

C. CCA-Based Objective

The CCA-based objective aims to learn the correlation between network structure distribution and node attribute distribution. First, we sample the structure embedding $\mathbf{Z}_h$ and the node attribute embedding $\mathbf{Z}_f$ from the distribution of network structure $q(\mathbf{Z}_h|\mathbf{X}, \mathbf{A})$ and the distribution of node attributes $q(\mathbf{Z}_f|\mathbf{X}, \mathbf{I})$, and normalize the node embedding of the two views as follows:

$$\begin{aligned} \mathbf{Z}'_h &= \frac{\mathbf{Z}_h - \mu(\mathbf{Z}_h)}{\sigma(\mathbf{Z}_h) * N^{\frac{1}{2}}} \\ \mathbf{Z}'_f &= \frac{\mathbf{Z}_f - \mu(\mathbf{Z}_f)}{\sigma(\mathbf{Z}_f) * N^{\frac{1}{2}}}. \end{aligned} \quad (11)$$

Then, from (3), CaCo maximizes the correlation between the two view distributions by minimizing the invariance of the network structure embedding $\mathbf{Z}_h$ and the node attribute embedding $\mathbf{Z}_f$. The invariance loss $\mathcal{L}_{inv}$ is defined as

$$\mathcal{L}_{inv} = \|\mathbf{Z}'_h - \mathbf{Z}'_f\|_F^2. \quad (12)$$

To avoid collapsed solutions, the following decorrelation $\mathcal{L}_{dco}$ loss is proposed to ensure that the individual dimensions of the features are uncorrelated

$$\mathcal{L}_{dco} = \|\mathbf{Z}'_h \mathbf{Z}'_h{}^T - \mathbf{I}\|_F^2 + \|\mathbf{Z}'_f \mathbf{Z}'_f{}^T - \mathbf{I}\|_F^2. \quad (13)$$

The CCA-based objective is defined as follows:

$$\mathcal{L}_{CCA} = \mathcal{L}_{inv} + \lambda * \mathcal{L}_{dco} \quad (14)$$

where λ is the tradeoff between the two terms.

D. Loss Function and Anomaly Score

The training objective of CaCo is to optimize the CCA-based loss and the KL divergence of the network structure distribution

Algorithm 1: CaCo.

Require:

Attributed network $\mathcal{G} = \{\mathbf{A}, \mathbf{X}\}$; Identity graph $\mathcal{G} = \{\mathbf{A}, \mathbf{I}\}$; weight-sharing encoder f ; the prior distribution p and the hyperparameter λ of CCA-based objective.

- 1: **for** $epoch = 1$ to T **do**
 - 2: Generate the network structure distribution $q(\mathbf{Z}_h|\mathbf{H}, \mathbf{A})$ via (4) and (6) and generate the node attribute distribution $q(\mathbf{Z}_f|\mathbf{X}, \mathbf{I})$ via (5) and (6).
 - 3: Align the distribution of network structure and node attribute via (10).
 - 4: Sample the network structure embedding $\mathbf{Z}_h$ and node embedding $\mathbf{Z}_f$ from the distribution of network structure $q(\mathbf{Z}_h|\mathbf{H}, \mathbf{A})$ and node attributes $q(\mathbf{Z}_f|\mathbf{X}, \mathbf{I})$.
 - 5: Normalize the network structure embedding $\mathbf{Z}_h$ and node embedding $\mathbf{Z}_f$ by (11).
 - 6: Input the normalized network structure embedding $\mathbf{Z}'_h$ and node attribute embedding $\mathbf{Z}'_f$ into CCA-objective, and update the CaCo with its stochastic gradient by (15).
 - 7: **end for**
 - 8: Compute the correlation between network structure embedding $\mathbf{Z}'_h$ and node embedding $\mathbf{Z}'_f$ via (12) and detect the anomaly by it.
 - 9: **return** f .
-

and node attribute distribution

$$\mathcal{L} = \mathcal{L}_{CCA} + \mathcal{L}_{KL}. \quad (15)$$

Unlike other methods, the anomaly score of CaCo is defined as the correlation of two views. A strong correlation between the network structure and the node attribute is determined as normal; otherwise, it is determined as abnormal. The algorithm of CaCo is described as follows.

IV. EXPERIMENTS

A. Datasets

To evaluate the effectiveness of CaCo, we tested the CaCo methods on six public datasets, including four citation benchmark datasets (Cora, Citeseer, Pubmed, and ACM) and two social benchmark datasets (Flickr, BlogCatalog). Table I shows the statistics of these datasets. The details of these datasets are as follows.

- 1) *Citation Networks*: Cora, Citeseer, Pubmed,¹ and ACM are four publicly used citation network benchmark datasets with scientific publications as nodes and the citation links between these scientific publications as edges.
- 2) *Social Networks*: BlogCatalog and Flickr² are two typical social network benchmark datasets, where users of social

¹[Online]. Available: <http://linqs.cs.umd.edu/projects/projects/lbc>

²[Online]. Available: <http://socialcomputing.asu.edu/pages/datasets>

TABLE I
STATISTICS OF SIX DATASETS

Types	Dataset	$\mathcal{V}$	$\mathcal{E}$	$\mathcal{A}$	Class
Citation Network	Cora	2 708	5 429	1 433	7
	Citeseer	3 327	4 732	3 703	6
	Pubmed	19 717	44 338	500	3
	ACM	16 485	71 980	8 337	9
Social Network	Flickr	7 575	239 738	12 407	9
	BlogCatalog	5 196	121 743	8 189	6

The upper four datasets are citation networks and the remaining two datasets are social networks.

websites are represented as nodes and the following relationships between users are represented as edges.

Based on previous studies on these datasets [13], [14], we chose one of the classes as the normal class and the remaining classes as the anomalous class. In our experiments, we randomly sampled only 60% of the normal nodes for training, and the remaining 15% and 25% randomly selected the same number of anomalous nodes for validation and testing, respectively.

B. Experimental Settings

In this section, we introduce the settings of our experiments including baselines, experimental design, and metrics.

1) *Baseline*: We compared the proposed model CaCo with four types of baselines. *First*, the traditional one-class classification-based methods include OC-SVM [5] and Deep-SVDD [8], which detect the anomaly by a learned hypersphere containing normal data. In this article, we learn the hypersphere on the network structure and node attributes, respectively. *Second*, graph-embedding-based methods such as DeepWalk [20], GAE [21], ARGAs [22], Dominant [11], and ComGA [29] employ graph learning methods to extract graph information and use the reconstruction errors to detect the anomaly. *Third*, one-class GNN-based methods include OC-GNN [13] (OC-GCN, OC-SAGE) and Dual-SVDAE [14], which combine the powerful representation capabilities of GNNs along with the classical one-class objective to conduct anomaly detection on graph structure data. *Fourth*, graph SSL-based methods include CCA-SSG [15], which utilizes CCA to optimize the feature-level objective and Conad [30], which integrates the anomaly knowledge into the Siamese GNN encoder by contrastive loss and detects anomaly nodes via reconstruction errors.

2) *Experimental Design and Metric*: In the experiments, CaCo is trained with 500 iterations on all datasets and optimized by Adam with a learning rate of 0.001. The embedding dimension is set to 64 and the layers of GNN are set to 2 for all datasets. The hyperparameter λ is set to 1 for all datasets. For the baseline methods, we use the publicly available implementations from the original papers and set the parameters by grid search. To measure the performance of CaCo with baselines, similar to the previous studies [13], [14], in this article, we employ the widely used *AUC* score and *AP* score as metrics. All methods

are evaluated under five different random seeds and the mean performance and the mean variance were reported as the final results. In the experiments, CaCo is implemented using Pytorch³ and Deep Graph Library.⁴ All experiments were conducted on an Ubuntu 16.04 Server with 128 GB of RAM, Intel(R)-Xeon(R)-Gold-5220R(2.20 GHz), and a GeForce RTX 3090 Graphics Card with 24 GB of RAM.

C. Experimental Results

The performance of the proposed model was evaluated quantitatively by comparing it with the state-of-the-art methods. The results of all methods on six datasets are shown in Table II, and the results of each method on six datasets by treating each class as a normal class are shown in Fig. 3.

In the experimental results, our proposed framework CaCo achieves the best performance on six datasets. CaCo obtains a significant improvement of 5.90% on AUC and 6.43% on AP compared to the best results in the baseline. The main reason is that CaCo captures the distributional relationships on node attribute and network structure. In comparison with the traditional anomaly detection methods, OCSVM and Deep-SVDD perform worse than the other methods. The reason is that they cannot simultaneously capture the information of node attributes and complex network structures. The performance of graph-embedding-based methods is not satisfactory, because they effectively measure the anomaly via a single aspect (reconstruction error). For the method ComGA, it uses the community-aware method to learn the node embedding, which can be more easily separated in the latent space. However, it uses reconstruction error as anomaly score, which makes it impossible to detect abnormal nodes similar to normal nodes. The one-class GNN-based methods, such as OC-GCN and Dual-SVDAE, do not show competitive performance, even though they are designed for graph structure feature extraction and hypersphere learning for anomaly detection. This lies in the inability of the one-class method to measure abnormal nodes whose latent space is similar to the normal nodes. Graph self-supervised methods, such as Conad, perform worse than CaCo, due to the fact that it learns the graph embedding using prior human knowledge for data augmentation and cannot include all exception patterns. This makes it unable to detect anomaly nodes that have similar patterns to normal nodes. Unlike other baselines, CaCo aims to capture the correlations between network structure and node attributes, and when a node is abnormal, the network embedding will be different from the node attribute embedding while other baselines aim to learn the representation of nodes in the latent space, ignoring the relationship between network structure and node attributes. This is the core idea of CaCo, and the results proved it. Besides, the distribution alignment makes CaCo capture the different information between network structure and node attributes, and we can obtain the same conclusion in the ablation study.

³[Online]. Available: <https://pytorch.org/>

⁴[Online]. Available: <https://www.dgl.ai/>

TABLE II
ANOMALY DETECTION PERFORMANCE

Methods	Cora		Citeseer		Pubmed	
	AUC	AP	AUC	AP	AUC	AP
OCSVM [5]	60.92±5.41	57.51±4.48	55.42±3.79	54.74±2.67	54.62±9.67	53.89±7.16
Deep-SVDD(Attr) [6]	63.81±3.32	62.21±3.50	67.18±7.33	66.73±7.82	70.77±9.44	71.03±10.64
Deep-SVDD(Stru) [6]	70.13±4.51	69.21±5.30	73.47±10.23	73.70±9.74	71.39±9.77	71.62±11.02
DeepWalk [5]	66.83±12.82	64.73±11.86	50.37± 12.68	51.84± 8.12	52.40±8.40	52.35±6.16
GAE [21]	56.49±7.46	59.23±7.75	52.28±7.43	52.63±5.42	53.88±6.21	53.26±4.44
ARGA [22]	75.18±8.87	73.14±8.98	71.42±5.31	70.08±4.88	53.74±3.53	52.99±2.85
Dominant [11]	87.29±3.71	78.35±6.23	80.17±7.78	72.36±6.15	65.29±2.21	61.66±2.47
OC-GCN [13]	68.19±11.19	66.76±9.15	75.29±6.55	73.29±6.35	66.65±5.37	68.05±6.22
OC-SAGE [13]	90.99±2.60	89.37±3.12	84.13±6.67	81.52±5.53	74.60±6.17	75.11±6.83
Dual-SVDAE [14]	95.09±2.50	95.05±2.66	86.54±6.72	86.20±7.08	91.32±2.37	92.39±2.55
CCA-SSG [15]	89.40±5.78	82.01±9.35	81.86±7.69	73.58±7.00	69.83±11.49	62.58±8.45
Conad [30]	52.63±2.44	53.73±2.82	52.80±3.82	54.42±4.61	53.54±5.53	53.27±4.36
ComGA [29]	93.60±2.39	92.25±2.72	85.69±5.84	84.64±5.69	55.29±2.70	54.79±2.42
CaCo	97.62±1.04	97.59±1.03	92.73±4.52	91.97±5.42	92.62±2.29	93.04±2.64
Methods	ACM		Flickr		BlogCatalog	
	AUC	AP	AUC	AP	AUC	AP
OCSVM [5]	52.47±9.47	54.03±8.90	51.51±12.12	53.54±9.55	53.17±7.30	54.08±5.88
Deep-SVDD(Attr) [6]	64.59±2.28	63.56±3.04	76.59±4.27	73.19±4.91	60.08±2.95	57.53±2.23
Deep-SVDD(Stru) [6]	70.30±7.22	69.72±7.91	70.24±7.77	67.68±9.75	70.90±6.70	71.55±8.65
DeepWalk [5]	57.80±16.36	59.26±12.46	53.06±12.59	56.19±12.62	55.89±12.68	58.74±13.01
GAE [21]	55.30±5.66	55.70±5.66	52.31±11.82	53.85±7.73	61.54±4.77	65.25±6.31
ARGA [22]	92.36±9.86	92.33±9.94	68.87±6.16	70.22±5.16	83.66±2.12	84.91±2.10
Dominant [11]	91.32±10.74	91.21±9.97	49.65±11.61	52.50±9.11	49.84±9.39	51.83±6.89
OC-GCN [13]	70.08±6.65	69.48±7.25	69.72±8.69	66.94±9.73	68.69±5.51	69.16±7.86
OC-SAGE [13]	78.03±6.41	73.13±5.52	66.32±5.65	64.36±5.93	65.85±4.44	65.02±3.82
Dual-SVDAE [14]	91.33±5.45	91.01±6.11	69.54±8.69	66.80±9.88	85.52±3.87	87.11±3.84
CCA-SSG [15]	87.46±5.48	80.73±5.83	68.91±8.63	62.61±7.30	87.46±5.48	80.73±5.83
Conad [30]	51.23±5.45	50.91±5.92	52.62±9.40	55.10±9.05	56.41±8.39	58.89±9.49
ComGA [29]	–	–	57.88±10.23	58.32±9.58	68.25±8.67	68.15±7.98
CaCo	94.29±2.89	94.69±2.89	85.13±6.32	86.87±5.18	92.34±3.38	92.97±3.68

All values are percentages (%) and averaged with StdDevs over five seeds. The best results are marked in bold.

Fig. 3 shows the AUC scores of all methods by considering each node class as a normal class on the six datasets. From the results, we find that CaCo performs more effectively and robustly than the other baselines. This is due to the CaCo capturing network structure and node attribute information jointly by aggregating node identities using identity graphs, weight sharing GCN to map network structure and node attribute information to the same distribution. In particular, the performance of the baseline methods is worse than that of the other classes in the “7” class of Cora, the “1” class of Citeseer, and the “5” class of BlogCatalog datasets. The main reason behind this may be the relatively small size of these classes in these datasets, respectively. The “7” class of Cora is only 12.96%, the “1” class of Citeseer is only 7.35%, and the “2” class of BlogCatalog is only 14.95%, which makes these methods unable to have enough training for those classes to distinguish them from others. Unlike the baseline methods, CaCo has the weight-sharing GCN encoder and identity graph for learning the information about the network structure and node attribute of normal nodes, which

helps CaCo to capture the complete semantic information of normal data. Besides, the correlation of two views as anomaly scores helps CaCo to effectively distinguish normal classes from other classes.

D. Parameter Study

In this section, we study the impact of different dimensions of node embedding, and the number of layers of the weight-sharing encoder, with the results being run on the Cora dataset, as shown in Fig. 4. We can notice that CaCo performs stably at the different dimensions of node embedding. This is due to the fact that CaCo captures key information about the network structure and node attributes used for anomaly detection. We also see that the performance of CaCo is stable as the number of layers increases due to the weight-sharing encoder using the identity graph as inputs, which enables CaCo to retain more node information in the feature learning process.

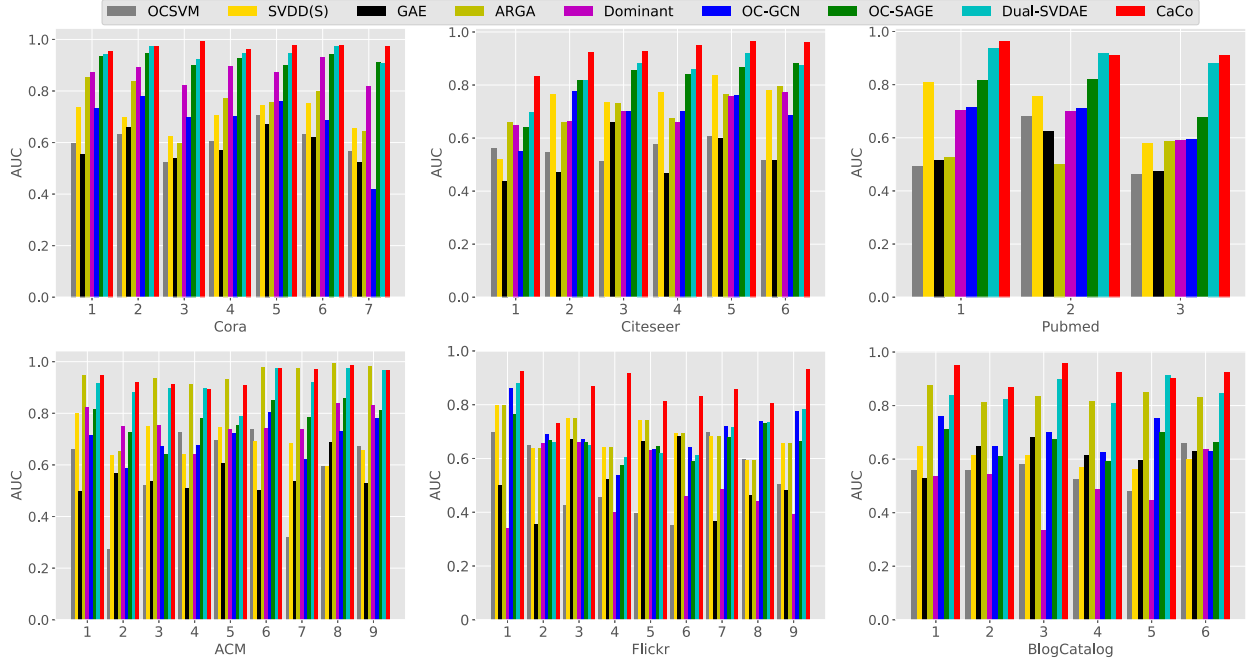


Fig. 3. Evaluation performance of CaCo and baselines by taking each class of nodes as a normal class on six datasets. The abscissa represents the classes and the ordinate represents the AUC score.

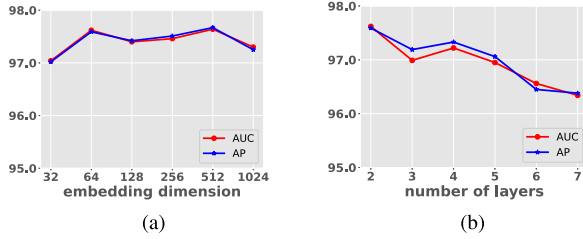


Fig. 4. Impact of different dimensions of embedding and the number of layers for weight sharing encoder on the Cora dataset.

E. Ablation Study

In this section, we compare CaCo with three variants on six datasets to study the different modules of CaCo used for anomaly detection. The variants are defined as follows.

- 1) **wo-dis**: This variant replaces distribution learning with feature learning to learn the correlation between network structure and node attributes.
- 2) **wo- $\mathcal{L}_{inv}$** : This variant studies the effectiveness of the invariant term by removing it from the loss function.
- 3) **wo- $\mathcal{L}_{dco}$** : Similar to wo- $\mathcal{L}_{inv}$, this variant studies the effectiveness of the decorrelation term by removing it from the loss function.

The results of the ablation study are shown in Table III. We can see that wo-dis performs worse than the others because learning the correlation between attribute distribution and structure distribution captures information about nodes that are not present in the training set while the distribution learning module captures key information for anomaly detection.

TABLE III
ABLATION STUDY OF CACO

Methods	Cora		Citeseer	
	AUC	AP	AUC	AP
wo-dis	49.89 $\pm$ 1.41	50.65 $\pm$ 1.22	51.18 $\pm$ 5.26	51.29 $\pm$ 3.63
wo- $\mathcal{L}_{inv}$	96.39 $\pm$ 1.76	96.48 $\pm$ 1.53	92.33 $\pm$ 4.92	91.63 $\pm$ 5.72
wo- $\mathcal{L}_{dco}$	50.31 $\pm$ 2.48	51.54 $\pm$ 1.66	52.92 $\pm$ 4.82	52.02 $\pm$ 2.16
CaCo	97.62$\pm$1.04	97.59$\pm$1.03	92.73$\pm$4.52	91.97$\pm$5.42
Methods	Pubmed		ACM	
	AUC	AP	AUC	AP
wo-dis	71.89 $\pm$ 4.75	71.65 $\pm$ 5.09	50.24 $\pm$ 0.93	50.19 $\pm$ 1.04
wo- $\mathcal{L}_{inv}$	94.21 $\pm$ 3.44	94.62 $\pm$ 3.07	92.58 $\pm$ 2.81	92.72 $\pm$ 3.17
wo- $\mathcal{L}_{dco}$	54.89 $\pm$ 6.07	56.02 $\pm$ 5.26	54.75 $\pm$ 3.00	52.66 $\pm$ 3.61
CaCo	94.29$\pm$2.89	94.69$\pm$2.89	92.62$\pm$2.29	93.04$\pm$2.64
Methods	Flickr		BlogCatalog	
	AUC	AP	AUC	AP
wo-dis	67.45 $\pm$ 3.61	68.39 $\pm$ 2.21	67.89 $\pm$ 1.56	69.65 $\pm$ 1.34
wo- $\mathcal{L}_{inv}$	80.23 $\pm$ 11.77	81.98 $\pm$ 11.05	88.56 $\pm$ 4.05	90.08 $\pm$ 4.14
wo- $\mathcal{L}_{dco}$	52.75 $\pm$ 3.15	51.96 $\pm$ 2.17	53.10 $\pm$ 4.36	51.52 $\pm$ 2.39
CaCo	85.13$\pm$6.32	86.87$\pm$5.18	92.34$\pm$3.38	92.97$\pm$3.68

All values are percentages (%) and averaged with StdDevs over five seeds. The best results are marked in bold.

For wo- $\mathcal{L}_{inv}$, we found that using only the decorrelation term does not yield good results on Flickr and BlogCatalog, but has good performance on the first four datasets, as shown in Fig. 5. Specifically, for the first four datasets, the distribution of anomaly scores have a clear dividing line, which means that those datasets are themselves strongly correlated with normal

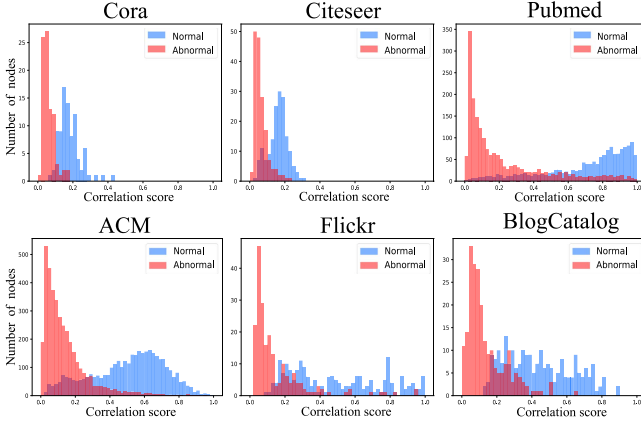


Fig. 5. Correlation score of $\text{wo-}\mathcal{L}_{\text{inv}}$ on six datasets.

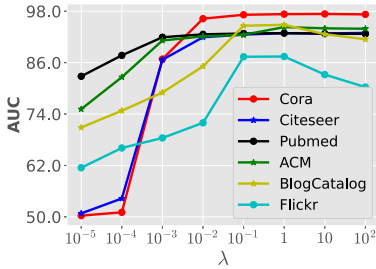


Fig. 6. Result of different values of tradeoff hyperparameter λ .

points and weakly correlated with abnormal points after optimization with the decorrelation term, but this is not the case for the Flickr and BlogCatalog datasets. Thus, for the first four datasets, CaCo can also perform well without the invariant term. For the Flickr and BlogCatalog datasets, CaCo needs to further optimize the invariant term.

For $\text{wo-}\mathcal{L}_{\text{dco}}$, it performs worse than the other algorithms, because, without decorrelation terms, CaCo will achieve collapsed solutions [15]. As shown in Fig. 6, we also study the impact of the decorrelation term on anomaly detection by setting the tradeoff hyperparameter λ from 10^{-5} to 100. We discovered that when λ is too small, the decorrelation term does not work, allowing CaCo to get collapsed solutions. When the value of λ is too large, we notice that the performance of CaCo is degraded, especially on Flickr and BlogCatalog. This is because the invariant term will be neglected. Therefore, the value of the hyperparameter λ is associated with the correlation between network embedding and node embedding. From Fig. 6, a value of the hyperparameter λ in $[0.1, 1]$ can achieve the optimal performance. This means that the performance of CaCo benefits from the invariance term. In contrast to $\text{wo-}\mathcal{L}_{\text{dco}}$, CaCo utilizes the decorrelation term to prevent solution collapse. Above all, the results demonstrate that the modules of CaCo can be used for attributed network anomaly detection.

F. Computational Analysis

In this section, we compare the complexity of CaCo with other baselines, as shown in Table IV. We have compared the

TABLE IV
COMPUTATION ANALYSIS FOR CaCo WITH OTHER METHODS

Methods	Params/MB	FLOPs/G
GAE [21]	3.8973	1.0058
ARGA [22]	1.2914	1.0093
Dominant [11]	3.8973	1.0058
OC-GCN [13]	1.3046	0.4312
OC-SAGE [13]	102.6528	4986.2418
Dual-SVDAE [14]	2.6197	126.75
CCA-SSG [15]	1.3046	0.3111
Conad [30]	1.9425	0.4951
ComGA [29]	196.4977	9543.0531
CaCo	1.321	0.4415

number of parameters and FLOPs of CaCo with other baselines. We can find that the number of parameters and FLOPs is smaller than most baselines. The reason is that CaCo uses only l -layers GCN as an encoder, a light-weighted network for graph data, which is simpler and more effective to operate compared to other networks such as GraphSAGE. For ComGA, it first learns the node embedding of the community, which makes it necessary to design more modules and greatly increases the number of parameters. In addition, for GAE, dominant, and dual-SVDAE, the dual-encoder or decoder also requires a new network module, which makes them require more parameters and computations. CCA-SSG and Conad are similar to CaCo, except that they use l -layers GCN to capture the information of attributed networks, which makes them reduce the number of parameters and computations.

G. Visualization

In this section, we visualize the embedding and correlation scores class “1” of six datasets. Fig. 7 visualizes the node embedding of the test nodes learned by ARGAs, Dual-SVDAE, and CaCo, respectively, with *blue* points representing normal nodes and *orange* points representing the abnormal nodes. We can see that CaCo, which separates normal nodes from abnormal nodes, is better than the above methods. The reason is that CaCo aligns the distribution of network structures and node attributes in the latent space and utilizes the correlation between the distributions of two views as an objective to detect the anomaly. Furthermore, we notice that the distribution of nodes in the latent space is not as tight as other methods, which is due to the fact that CaCo focuses on the separation of normal and abnormal nodes, rather than methods such as OC-GNN and Dual-SVDAE that focus on learning a compact hypersphere of normal nodes to separate normal nodes from abnormal nodes. Fig. 8 shows the correlation scores of the six datasets, we can find that the anomaly nodes have smaller correlation values while the normal nodes have larger correlation values. The results indicate the effectiveness of the proposed CaCo.

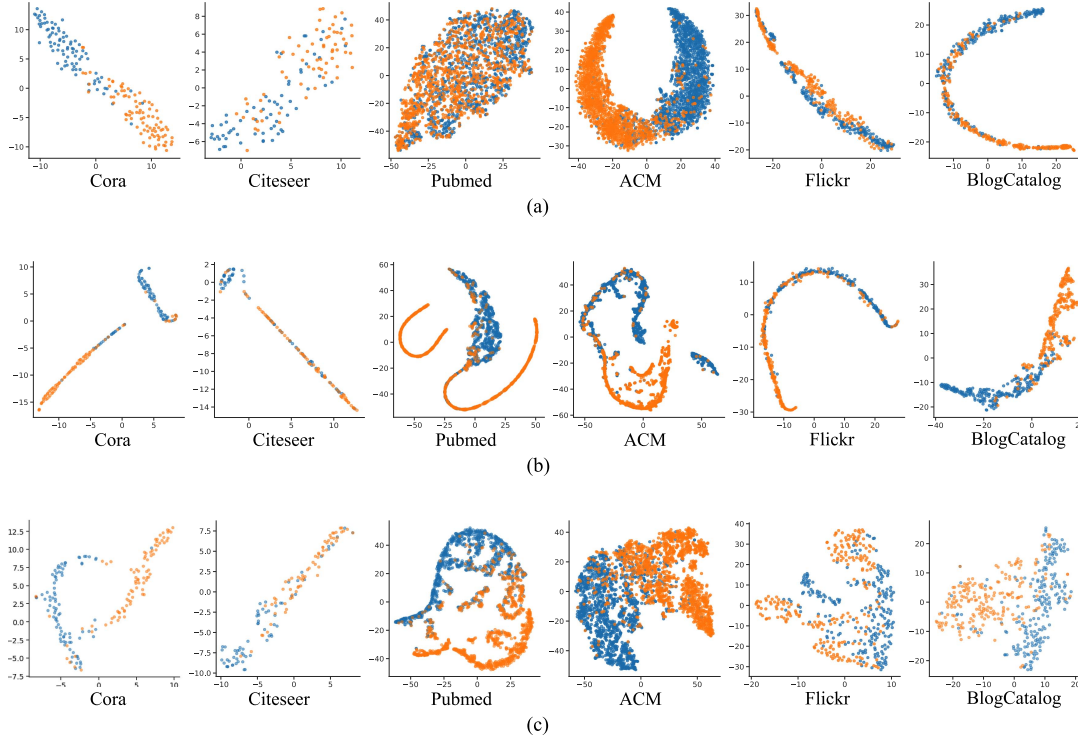


Fig. 7. T-SNE visualization of node embedding derived by different models on six datasets. *Blue* points represent the normal nodes and *orange* points represent the abnormal nodes respectively. (a) ARGA. (b) Dual-SVDAE. (c) CaCo.

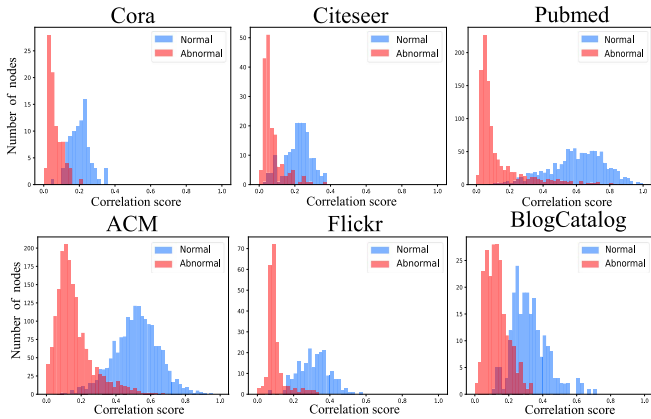


Fig. 8. Correlation score of CaCo on six datasets.

V. CONCLUSION

In this article, we propose a CCA-based method for attributed network anomaly detection (CaCo), which captures the correlation between node attributes and the network structure of attributed networks for anomaly detection. Specifically, CaCo designs weight-sharing GCN as an encoder to learn the distribution network structure and node attributes, and we align them to a sharing latent space via KL divergence. Then, the CCA objective is designed to maximize the correlation of normal nodes in terms of network structure and node attributes, and the anomaly score is defined as the correlation of two views to detect the anomaly points. Extensive experiments on six real-world

datasets demonstrate the effectiveness of the proposed method compared to the state-of-the-art techniques. In the future, we will focus on the correlation between the global and the local nodes to detect the anomaly points.

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